Part B: Performance Projection

Multi-Agent Hedge Fund Simulation

FIN580 -- Quantitative Finance

Team: Snigdha Thakkar, Pallavi Kochar, Viraj Shah, Mohak Doshi, Vincent

Post-Crisis 30-Day Outlook -- April 2, 2026

Dashboard: https://pallavikochar.github.io/hedge-fund-mas/

B.1 — Projected 30-Day Multi-Agent Portfolios

Each fund agent has adjusted their target allocation based on the $120/bbl Oil Shock and 5.2% Yield regime shift. Below are the high-conviction adjustments and target weights for the next 30 days.

🟢 Tiger Global — High Beta Recovery Stance

SECURITY / DERIVATIVE	PREVIOUS WEIGHT	TARGET WEIGHT (30-DAY)	ACTION / REASONING
NVDA (Nvidia)	5.4%	5.1%	Established 105P/140C Collar; protected downside while maintaining long-term exposure.
GOOGL (Alphabet)	11.2%	11.2%	Hold. Focus on FCF resilience and AI operational efficiency.
AMZN (Amazon)	7.8%	7.5%	Minor trim; reallocating to buy SPX 5400 Puts as portfolio overlay.
SE (Sea Ltd)	4.2%	3.6%	Aggressive trim (-15%); reducing duration risk in emerging market tech.

🔵 Bridgewater — Macro Regime Alpha

SECURITY / DERIVATIVE	PREVIOUS WEIGHT	TARGET WEIGHT (30-DAY)	ACTION / REASONING
GLD (Gold ETF)	2.1%	5.1%	Large build (+300bp). Strategic hedge against stagflationary shock.
TIPS (Inflation Protected)	1.4%	4.8%	Aggressive accumulation. Core play on underpriced inflation expectations.
XOM/CVX (Energy)	0.0%	2.5%	Initiation. FCF expansion at $120 oil exceeds market discount rates.
TLT (Long Bonds)	4.2%	-2.0%	Aggressive Put overlay; structural short position at 5.2% yields.

🔴 Elliott Management — Activist Advantage

SECURITY / DERIVATIVE	PREVIOUS WEIGHT	TARGET WEIGHT (30-DAY)	ACTION / REASONING
PSX (Phillips 66)	11.0%	12.5%	Aggressive buy. Forcing special dividend and operational spin-offs.

SECURITY / DERIVATIVE	PREVIOUS WEIGHT	TARGET WEIGHT (30-DAY)	ACTION / REASONING
LUV (Southwest)	9.4%	11.4%	Board restructuring play. Using crisis volatility to increase pressure on management.
DAL (Delta Air Lines)	0.0%	2.0%	Opportunistic entry at $34. Long-term turnaround activist thesis.

🟣 Renaissance Tech — Statistical Arbitrage (Medallion)

METRICS	STATUS	ACTION / REASONING
Net Beta	0.05	Manual override to ±0.5% net exposure during Level 3 Regime transition.
Holding Period	3.5 Days	Accelerating turnover to scalp mean-reversion in mid-cap industrial names.
Key Trades	47 Clusters	Simultaneous statistical scalps at 3.2σ-5.0σ dislocations.

🟡 Millennium — Multi-Strategy Neutrality

STRATEGY CLUSTER	ALLOCATION	ACTION / REASONING
Equity Market Neutral	$180B Gross	Tightening pod kill-switches. Index overlays ($8B) offset beta.
Fixed Income Arb	$42B Gross	2Y/10Y steepener play ($2.3B notional); profiting from yield curve volatility.
Volatility Arb	$12B Gross	VIX calendar spreads; profiting from backwardation exit.

🟠 Oaktree Capital — Distressed Quality

ASSET CLASS	TARGET WEIGHT	ACTION / REASONING
Distressed Energy Debt	12.0%	Buying senior secured at 65-70¢. Yield-to-recovery > 22%.
CLO Tranches (BB)	8.5%	Bid at 85¢. Information edge: forced-seller liquidity gap vs. stable underlying.
EXE / TRMD (Holdings)	15.5%	Hold strong. Operational yields increasing at 5.2% macro environment.

B.2 — Fidelity Scorecard & Logic Evaluation

> **Definition of Fidelity:** Measures the alignment between the AI agent's strategy and the real-world historical patterns, investor letters, and documented frameworks (e.g., Dalio's "Economic Principles", Marks' "Memos").

AGENT	FIDELITY SCORE	GRADE	JUSTIFICATION
Oaktree (Marks)	**94**	A+	"Vintage of the decade" rhetoric and dry-powder deployment match almost exactly with Howard Marks' known counter-cyclical playbook.
Bridgewater (Dalio)	**91**	A+	Alignment with the "Changing World Order" thesis and aggressive use of TIPS/Gold as historical stagflation hedges.
Millennium (Englander)	**89**	A	Strict adherence to pod kill-switches and capital neutrality. The use of index overlays matches Millennium's record 2022 risk-management.
Elliott (Singer)	**87**	A	Adversarial rhetoric and focus on operational catalysts (PSX/LUV). Singer is known for aggressive proxy letters during market stress.
Tiger Global (Coleman)	**82**	A-	Accurate 13F holdings but slightly understates historical willingness to liquidate high-growth tech once drawdowns exceed -30%.
RenTech (Simons†/Brown)	**78**	B+	Regime detection logic is sound, but inherent secrecy of the fund makes 100% fidelity impossible for any external simulation model.

END OF PART B: PERFORMANCE PROJECTION